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Studierichting: Informatica
Oefeningenreeks 1C nr. 45.9

Trek de derdemachtswortel in $\mathbb{C}$:

$z^3 = 1 - i$

$r = \sqrt{2}$ en $\theta = \text{Bgtan}(-1) = \dfrac{3\pi}{4} + \pi = \dfrac{7\pi}{4}$

$z^3 = \sqrt{2}\,\text{cis}\left(\dfrac{\frac{7\pi}{4} + k2\pi}{3}\right)$ met $k \in \{0, 1, 2\}$

$(k = 0)$ $z_1 = \sqrt[6]{2}\,\text{cis}\left(\dfrac{7\pi}{12}\right)$

$(k = 1)$ $z_2 = \sqrt[6]{2}\,\text{cis}\left(\dfrac{5\pi}{4}\right) = \sqrt[6]{2}\left(\dfrac{-\sqrt{2}}{2} - \dfrac{\sqrt{2}}{2}i\right) = \dfrac{-\sqrt[3]{4}}{2} - \dfrac{\sqrt[3]{4}}{2}i$

$(k = 2)$ $z_3 = \sqrt[6]{2}\,\text{cis}\left(\dfrac{23\pi}{12}\right)$

References